Analyzing E-commerce Consumer Behaviour

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```r
# Load libraries
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.0     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.2     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.1
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
```

```r
library(corrplot)
```

```
corrplot 0.95 loaded
```

```r
library(viridis)
```

```
Loading required package: viridisLite
```

```r
library(scales)
```

```
Attaching package: 'scales'
```

```
The following object is masked from 'package:viridis':

    viridis_pal

The following object is masked from 'package:purrr':

    discard

The following object is masked from 'package:readr':

    col_factor
```

```r
library(car)
```

```
Warning: package 'car' was built under R version 4.5.3

Loading required package: carData

Warning: package 'carData' was built under R version 4.5.3


Attaching package: 'car'

The following object is masked from 'package:dplyr':

    recode

The following object is masked from 'package:purrr':

    some
```

```r
library(pROC)
```

```
Warning: package 'pROC' was built under R version 4.5.3

Type 'citation("pROC")' for a citation.

Attaching package: 'pROC'

The following objects are masked from 'package:stats':

    cov, smooth, var
```

```
# --- Clean and Preprocess Data ---
df_clean <- read.csv("online_shoppers_intention.csv") %>%
  mutate(
    # Convert Revenue to Factor
    Revenue = factor(Revenue, levels = c(FALSE, TRUE), labels = c("No Purchase", "Purchase")

    # Convert Month (Define Order: Feb is the Reference Level)
    Month = factor(Month, levels = c("Feb", "Mar", "May", "June", "Jul", "Aug", "Sep", "Oct"

    # Convert VisitorType and Weekend
    VisitorType = as.factor(VisitorType),
    Weekend = factor(Weekend, levels = c(FALSE, TRUE), labels = c("Weekday", "Weekend")),

    # Create Numeric Label for calculations
    is_purchase = as.numeric(Revenue == "Purchase")
  ) %>%
  drop_na()

# Define a consistent theme for the project
project_theme <- theme_minimal() +
  theme(
    plot.title = element_text(face = "bold", size = 14),
    axis.title = element_text(size = 12),
    legend.position = "bottom"
  )
```

Exploratory Data Analysis (EDA) distribution of Page Values We begin by examining our primary predictors. Our initial hypothesis suggests that PageValues significantly influences the likelihood of a transaction.

```
# Check for missing values
cat("Missing values in dataset:", sum(is.na(df_clean)), "\n")
```

```
Missing values in dataset: 0
```

```
# View the distribution of our target variable: Revenue
table(df_clean$Revenue)
```

```

No Purchase    Purchase 
      10422        1908 
```

```r
ggplot(df_clean, aes(x = Revenue, y = PageValues, fill = Revenue)) +
  geom_boxplot(alpha = 0.7) +
  scale_fill_manual(values = c("#FF6666", "#00CC66")) +
  labs(
    title = "Does Page Value Predict Purchase?",
    subtitle = "Higher PageValues are strongly associated with successful transactions",
    x = "Transaction Outcome (Revenue)",
    y = "Page Value Score"
  ) +
  project_theme
```

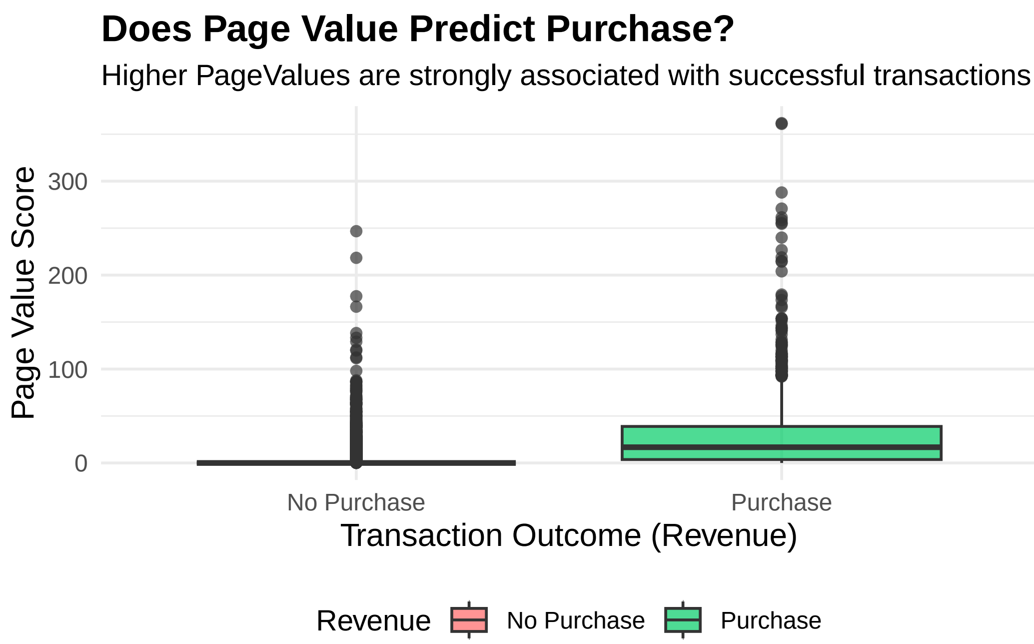


Figure 1: Comparison of Page Values for Purchase vs. No Purchase

```r
df_clean %>%
  group_by(Month) %>%
  summarise(Conversion_Rate = mean(Revenue == "Purchase")) %>%
  ggplot(aes(x = Month, y = Conversion_Rate, group = 1)) +
  geom_line(color = "steelblue", size = 1) +
  geom_point(size = 3, color = "darkblue") +
  scale_y_continuous(labels = percent) +
  labs(
    title = "E-commerce Seasonality: Monthly Conversion Rates",
```

```
    x = "Month of the Year",
    y = "Conversion Rate (%)"
  ) +
  project_theme
```

```
Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
i Please use `linewidth` instead.
```

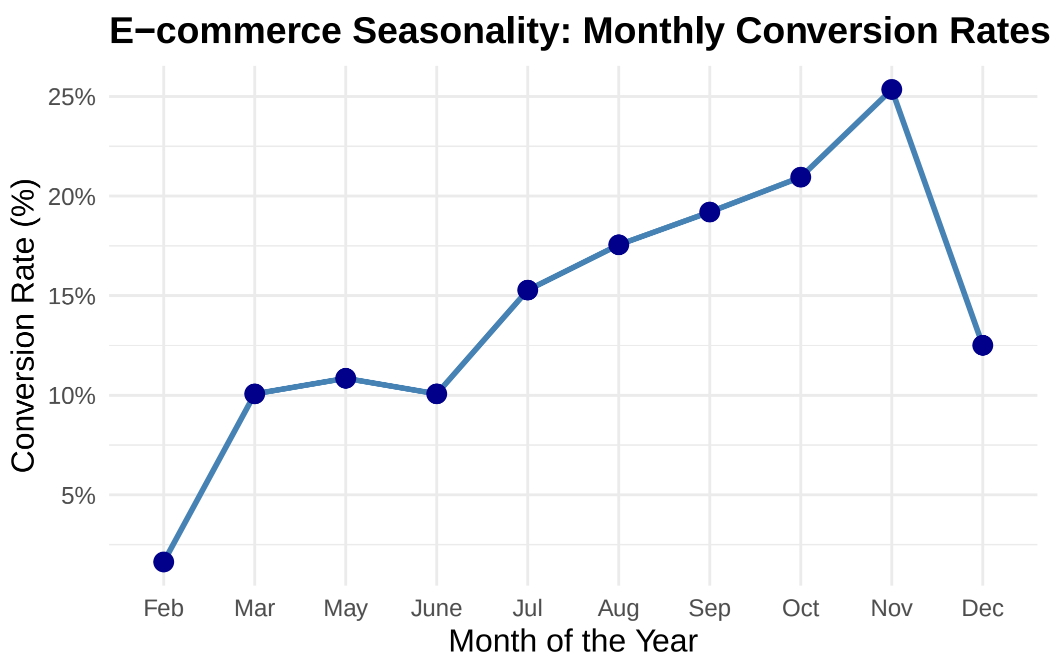


Figure 2: Monthly Conversion Rates Throughout the Year

```
ggplot(df_clean, aes(x = VisitorType, fill = Revenue)) +
  geom_bar(position = "fill") +
  scale_y_continuous(labels = percent) +
  labs(
    title = "Visitor Behaviour: New vs. Returning",
    x = "Visitor Category",
    y = "Percentage",
    fill = "Purchased?"
  ) +
  project_theme
```

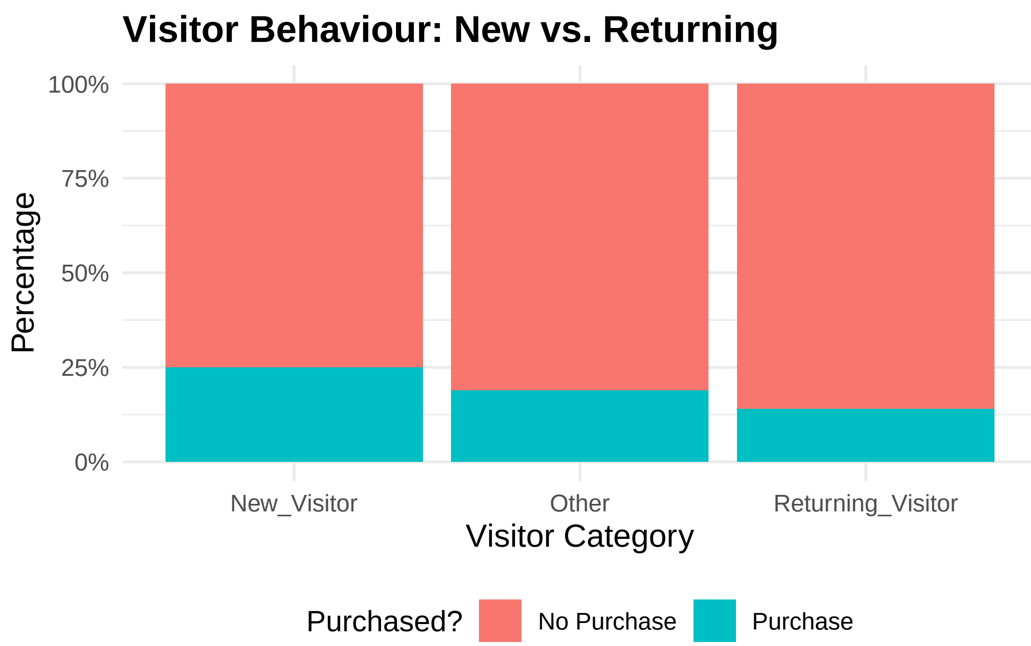


Figure 3: Purchase Proportion by Visitor Type

```
# Select numeric columns and calculate correlation
numeric_df <- df_clean %>% select(where(is.numeric))
cor_matrix <- cor(numeric_df)

corrplot(cor_matrix, method = "circle", type = "upper",
         tl.col = "black", tl.srt = 45,
         title = "Correlation Heatmap of Website Metrics",
         mar = c(0,0,1,0))
```

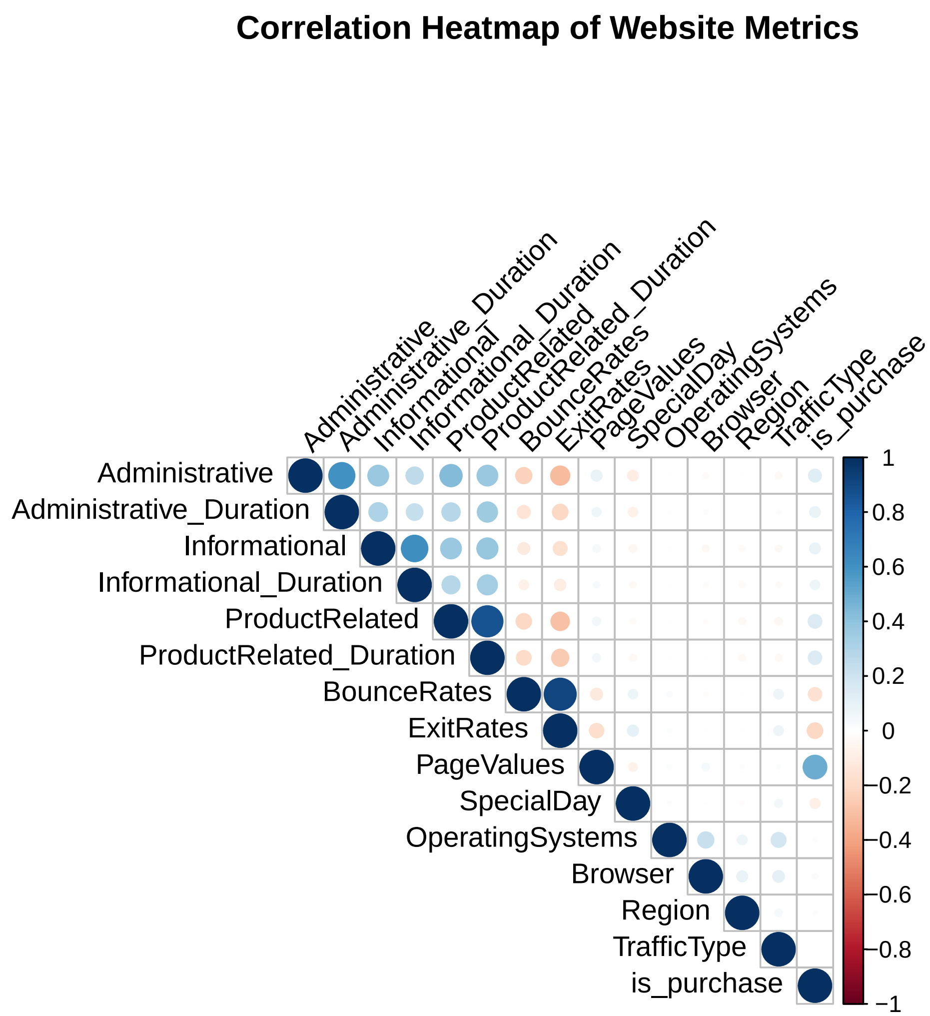


Figure 4

```
df <- read.csv("online_shoppers_intention.csv")

df_clean <- df %>%
  mutate(
    Revenue = factor(Revenue,
                     levels = c(FALSE, TRUE),
```

```r
                     labels = c("No Purchase", "Purchase")),

    Month = factor(Month,
                   levels = c("Feb", "Mar", "May", "June", "Jul", "Aug", "Sep", "Oct", "Nov"

    VisitorType = as.factor(VisitorType),
    Weekend = factor(Weekend,
                     levels = c(FALSE, TRUE),
                     labels = c("Weekday", "Weekend")),

    is_purchase = as.numeric(Revenue == "Purchase")
  ) %>%
  drop_na()

summary(df_clean)
```

```
 Administrative    Administrative_Duration Informational
 Min.   : 0.000   Min.   :   0.00         Min.   : 0.0000
 1st Qu.: 0.000   1st Qu.:   0.00         1st Qu.: 0.0000
 Median : 1.000   Median :   7.50         Median : 0.0000
 Mean   : 2.315   Mean   :  80.82         Mean   : 0.5036
 3rd Qu.: 4.000   3rd Qu.:  93.26         3rd Qu.: 0.0000
 Max.   :27.000   Max.   :3398.75         Max.   :24.0000

 Informational_Duration ProductRelated   ProductRelated_Duration
 Min.   :   0.00        Min.   :  0.00   Min.   :    0.0
 1st Qu.:   0.00        1st Qu.:  7.00   1st Qu.:  184.1
 Median :   0.00        Median : 18.00   Median :  598.9
 Mean   :  34.47        Mean   : 31.73   Mean   : 1194.7
 3rd Qu.:   0.00        3rd Qu.: 38.00   3rd Qu.: 1464.2
 Max.   :2549.38        Max.   :705.00   Max.   :63973.5

  BounceRates          ExitRates          PageValues        SpecialDay
 Min.   :0.000000   Min.   :0.00000   Min.   :  0.000   Min.   :0.00000
 1st Qu.:0.000000   1st Qu.:0.01429   1st Qu.:  0.000   1st Qu.:0.00000
 Median :0.003112   Median :0.02516   Median :  0.000   Median :0.00000
 Mean   :0.022191   Mean   :0.04307   Mean   :  5.889   Mean   :0.06143
 3rd Qu.:0.016813   3rd Qu.:0.05000   3rd Qu.:  0.000   3rd Qu.:0.00000
 Max.   :0.200000   Max.   :0.20000   Max.   :361.764   Max.   :1.00000

     Month       OperatingSystems    Browser           Region
 May    :3364   Min.   :1.000    Min.   : 1.000   Min.   :1.000
```

```
 Nov     :2998   1st Qu.:2.000    1st Qu.: 2.000   1st Qu.:1.000
 Mar     :1907   Median :2.000    Median : 2.000   Median :3.000
 Dec     :1727   Mean   :2.124    Mean   : 2.357   Mean   :3.147
 Oct     : 549   3rd Qu.:3.000    3rd Qu.: 2.000   3rd Qu.:4.000
 Sep     : 448   Max.   :8.000    Max.   :13.000   Max.   :9.000
 (Other):1337
  TrafficType               VisitorType       Weekend            Revenue
 Min.   : 1.00   New_Visitor      : 1694   Weekday:9462   No Purchase:10422
 1st Qu.: 2.00   Other            :   85   Weekend:2868   Purchase   : 1908
 Median : 2.00   Returning_Visitor:10551
 Mean   : 4.07
 3rd Qu.: 4.00
 Max.   :20.00

  is_purchase
 Min.   :0.0000
 1st Qu.:0.0000
 Median :0.0000
 Mean   :0.1547
 3rd Qu.:0.0000
 Max.   :1.0000
```

```r
n_total <- nrow(df_clean)
n_purchase <- sum(df_clean$Revenue == "Purchase")
n_no_purchase <- n_total - n_purchase

df_clean <- df_clean %>%
  mutate(case_weight = ifelse(Revenue == "Purchase",
                              n_total / (2 * n_purchase),
                              n_total / (2 * n_no_purchase)))
```

```r
# --- Step 1: Variable Selection (identify the optimal combination of predictors) ---
# Note: no weights passed here, keeping the environment clean for step()
predictors <- setdiff(names(df_clean), c("Revenue", "case_weight", "is_purchase"))
clean_formula <- as.formula(paste("Revenue ~", paste(predictors, collapse = " + ")))

# Fit a temporary unweighted model
temp_step_model <- glm(clean_formula, data = df_clean, family = binomial)

# Stepwise variable selection (trace = 0 keeps Render output clean)
aic_var_selection <- step(temp_step_model, direction = "both", trace = 0)
```

```
Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
```

```r
# --- Step 2: Final Weighted Model ---
# Extract the best-fitting formula from stepwise selection
final_formula <- formula(aic_var_selection)

# Refit with case weights and quasibinomial to suppress non-integer warnings
aic_model <- glm(final_formula,
                 data    = df_clean,
                 family  = quasibinomial,
                 weights = df_clean$case_weight)

# View final model results
summary(aic_model)
```

```
Call:
glm(formula = final_formula, family = quasibinomial, data = df_clean,
    weights = df_clean$case_weight)

Coefficients:
                              Estimate Std. Error t value Pr(>|t|)
(Intercept)                 -2.149e+00  1.877e+05       0        1
Informational                7.906e-02  8.224e+03       0        1
ProductRelated               2.113e-03  4.865e+02       0        1
ProductRelated_Duration      7.588e-05  1.195e+01       0        1
ExitRates                   -1.482e+01  5.133e+05       0        1
PageValues                   1.239e-01  1.472e+03       0        1
MonthMar                     1.090e+00  1.854e+05       0        1
MonthMay                     9.098e-01  1.844e+05       0        1
MonthJune                    1.492e+00  1.993e+05       0        1
MonthJul                     1.974e+00  1.908e+05       0        1
MonthAug                     1.831e+00  1.909e+05       0        1
MonthSep                     1.987e+00  1.900e+05       0        1
MonthOct                     1.752e+00  1.891e+05       0        1
MonthNov                     2.430e+00  1.839e+05       0        1
MonthDec                     1.236e+00  1.857e+05       0        1
OperatingSystems            -6.106e-02  1.360e+04       0        1
Browser                      3.516e-02  6.933e+03       0        1
VisitorTypeOther            -4.991e-01  1.916e+05       0        1
VisitorTypeReturning_Visitor -3.033e-01  3.264e+04       0        1
WeekendWeekend               9.472e-02  2.590e+04       0        1
```

```
(Dispersion parameter for quasibinomial family taken to be 216420485211)

    Null deviance: 17093  on 12329  degrees of freedom
Residual deviance: 10405  on 12310  degrees of freedom
AIC: NA

Number of Fisher Scoring iterations: 7
```

```r
# --- Step 1: Manually define predictors (exclude auxiliary columns) ---
# Ensure the model only sees raw features, not weight or helper columns
predictors <- setdiff(names(df_clean), c("Revenue", "case_weight", "is_purchase"))
clean_formula <- as.formula(paste("Revenue ~", paste(predictors, collapse = " + ")))

# --- Step 2: Fit base model (for variable selection only) ---
# No weights at this stage - keeps AIC logic clean and avoids errors in step()
step_model_base <- glm(clean_formula, data = df_clean, family = binomial)

# --- Step 3: Stepwise variable selection ---
# trace = 0 suppresses the selection log, keeping the rendered document clean
aic_var_selection <- step(step_model_base, direction = "both", trace = 0)
```

```
Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
```

```r
# --- Step 4: Fit final weighted model ---
# Extract the selected formula from aic_var_selection
final_formula <- formula(aic_var_selection)

# Key fix: use df_clean$case_weight to explicitly reference the column
aic_model <- glm(final_formula,
                 data    = df_clean,
                 family  = quasibinomial,
                 weights = df_clean$case_weight)

# View results
summary(aic_model)
```

```
Call:
glm(formula = final_formula, family = quasibinomial, data = df_clean,
    weights = df_clean$case_weight)
```

```
Coefficients:
                               Estimate Std. Error t value Pr(>|t|)
(Intercept)                  -2.149e+00  1.877e+05       0        1
Informational                 7.906e-02  8.224e+03       0        1
ProductRelated                2.113e-03  4.865e+02       0        1
ProductRelated_Duration       7.588e-05  1.195e+01       0        1
ExitRates                    -1.482e+01  5.133e+05       0        1
PageValues                    1.239e-01  1.472e+03       0        1
MonthMar                      1.090e+00  1.854e+05       0        1
MonthMay                      9.098e-01  1.844e+05       0        1
MonthJune                     1.492e+00  1.993e+05       0        1
MonthJul                      1.974e+00  1.908e+05       0        1
MonthAug                      1.831e+00  1.909e+05       0        1
MonthSep                      1.987e+00  1.900e+05       0        1
MonthOct                      1.752e+00  1.891e+05       0        1
MonthNov                      2.430e+00  1.839e+05       0        1
MonthDec                      1.236e+00  1.857e+05       0        1
OperatingSystems             -6.106e-02  1.360e+04       0        1
Browser                       3.516e-02  6.933e+03       0        1
VisitorTypeOther             -4.991e-01  1.916e+05       0        1
VisitorTypeReturning_Visitor -3.033e-01  3.264e+04       0        1
WeekendWeekend                9.472e-02  2.590e+04       0        1

(Dispersion parameter for quasibinomial family taken to be 216420485211)

    Null deviance: 17093  on 12329  degrees of freedom
Residual deviance: 10405  on 12310  degrees of freedom
AIC: NA

Number of Fisher Scoring iterations: 7
```

```r
# 1. Generate predicted probabilities using the selected model
predicted_probs <- predict(aic_model, type = "response")

# 2. Compute ROC curve and AUC
roc_obj <- roc(df_clean$Revenue, predicted_probs)
```

```
Setting levels: control = No Purchase, case = Purchase
```

```
Setting direction: controls < cases
```

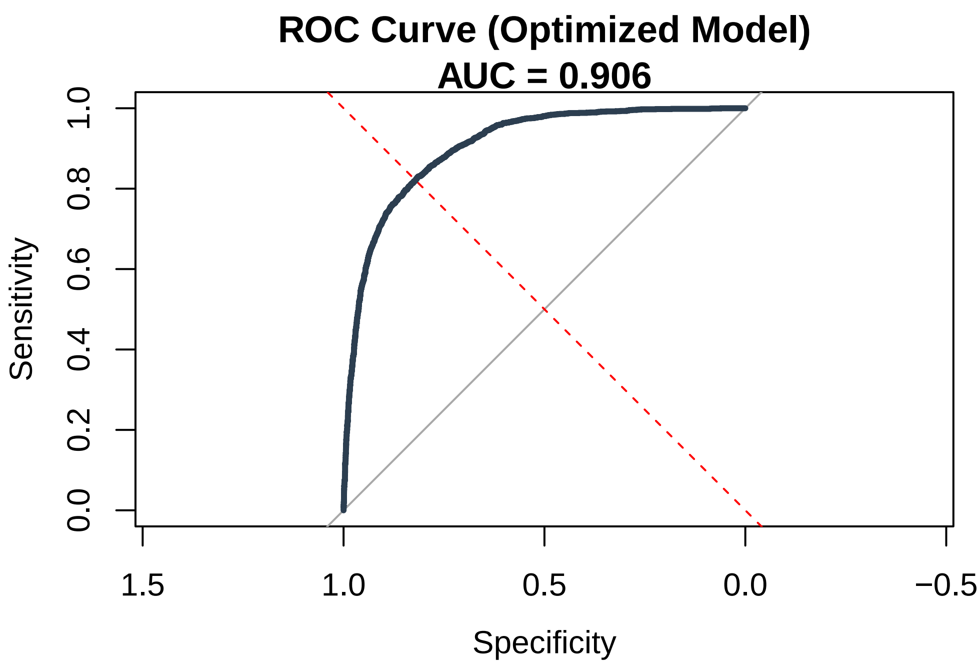
```r
auc_val <- auc(roc_obj)

# 3. Print key metrics
cat("AIC:", AIC(aic_model), "\n")
```

```
AIC: NA
```

```r
cat("AUC:", round(auc_val, 4), "\n")
```

```
AUC: 0.9065
```

```r
# 4. Plot ROC curve
plot(roc_obj, col = "#2c3e50", lwd = 3,
     main = paste("ROC Curve (Optimized Model)\nAUC =", round(auc_val, 3)))
abline(a = 0, b = 1, lty = 2, col = "red")
```



major variable

user intention data (PageValues, ExitRates)

time seryie data (Month)

user behavior data (ProductRelated_Duration, VisitorType)

“Excluded Variables”:

Technical: Browser, OS, Region (Minimal impact on intent).

Redundant: SpecialDay, Weekend (Captured by Month/Seasonality).

Operational: Administrative, Informational (Less predictive than product-related engagement).